Design criterion for lowering pressure pulsations

Index

1. Purpose & Scope 3
2. Definitions 3
3. General Procedure - Description 3
4. Influence parameters of pressure pulsations 5
5. Design guidelines for lowering pressure pulsations 7
6. Limitations 9
7. References 9
8. Attachment – Reduction of pressure pulsation on BB1 pump (job P30612) 10

1. Purpose & Scope

This procedure establishes the criteria to adopt for mitigating pressure pulsation effect on centrifugal pumps.

This procedure is applicable to all centrifugal pumps in with diffuser of volute casing with open and closed impeller.

This procedure applies to Nova Milanese plant and La Spezia plant.

2. Definitions

Abbreviations	Unit of Measurement/ equation (if applicable)	Description
n	[rpm]	Rotational speed of the pump rotor
Zimp		Number of impeller vanes
Zdiff		Number of diffuser vanes or volute tongues
Hst	[m or ft]	Pump head at best efficiency point
Pabs.st	[m^3/h or GPM]	Flowrate at best efficiency point
D2	[mm or in]	Maximum external diameter
LE		Leading edge of the vane/blade
TE		Trailing edge of the vane/blade
C3q	[m/s or ft/s]	Casing throat velocity at best efficiency point
C2	[m/s or ft/s]	Outlet impeller velocity
D3 or Dsw	[mm or in]	Diameter of LE diffuser or volute tongue cutwater
r	[kg/m^3]	Fluid density
Nq or Ns	(n*(Q.bep/suction sides)^0.5/Hst.bep^0.75)	Stage specific speed

3. General Procedure - Description

Pressure pulsations are a typical rotor stator interaction on centrifugal turbomachines. It consists of the fluctuation of the pressure field between the rotating and the static components.

A pressure pulsation can be defined as combinations of periodic signals, each of those is defined by the parameters listed below:

- Frequency is the number of fluctuations at a certain time. It is expressed in cycles per second (Hz) or cycles per minute (cpm). A graphical representation is presented in *Figure 1*.
- Amplitude is defined as the magnitude of the pressure signal. It can be measured as peak, peak to peak or rms (*Figure 1*). Typically, they are measured in Pa, mbar, psi or m.

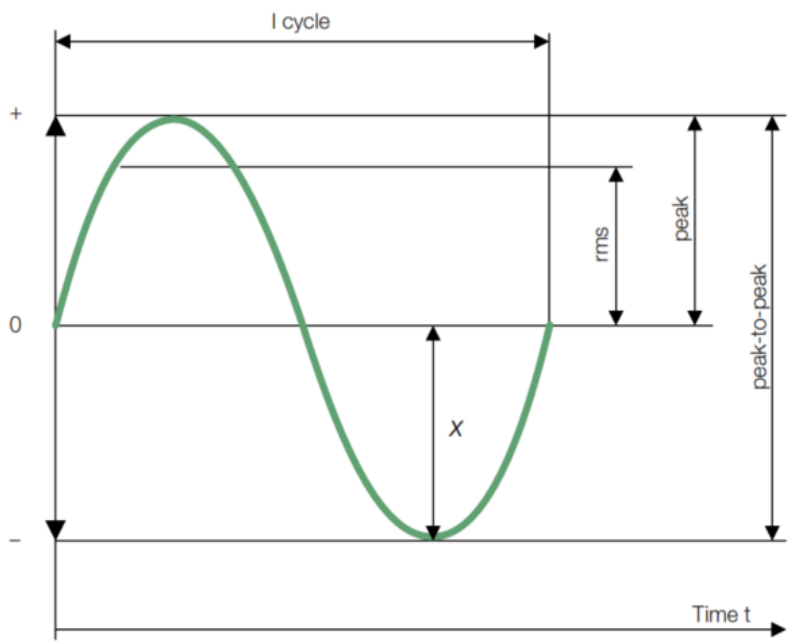


Figure 1: basics of pressure pulsations

Main sources of pressure pulsation are:

- Impingement of the wake forming downstream the trailing edge of rotating parts, on static components like diffusers or volute lips.
- Turbulent phenomena and vortical formations generated by flow separation and/or suction recirculation at part load.

The effect of the pressure pulsation is an alternating hydraulic fluctuation that interacts with the static parts and generates fatigue stresses on the components.

Usually, pressure pulsations have blade passing frequencies, defined as harmonics of the (Zimp x n/60).

If the blade pass frequency has a low separation margin with a pump/system natural frequency, usually of the bearing brackets, structural resonance can occur. In this condition, an increment of the effect of pressure pulsations is expected, causing failures or malfunctions (e.g., fatigue, excessive vibrations, etc.). Moreover, vane pass velocity fluctuations generate sound pressure waves, that travel in the pumped fluid and through the system at the speed of sound; in case of acoustic resonance, reflected waves can result in an amplified standing wave, causing damage and/or vibration problems. Both structural and acoustic resonance conditions must be avoided in the design phase, and the following technical considerations are in accordance with this hypothesis.

4. Influence parameters of pressure pulsations

A list of the main parameters that affect the pressure pulsations is shown below:

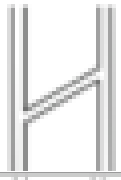
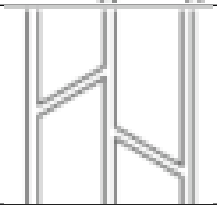
- Thickness and shape of trailing edge: the trailing edge shape plays a crucial role in the propagation of the wake from the impeller outlet to the static parts (diffuser or volute tongue).
- Vane loading at the impeller outlet: the blade loading on the impeller at the outlet is responsible for the velocity distribution on the impeller vanes. Low blade loading contributes to equalize the velocity distribution.
- Impeller inlet flow distribution: turbulent phenomena at the impeller's inlet, like vortex shedding from ribs, can cause pressure pulsations on the impeller leading edge.
- Impeller flow separation: at a high flow rate, the flow in the impeller starts separating due to incidence losses on the impeller's leading edge. The phenomenon is highly unsteady and generates fluctuations that can increase the pressure pulsation.
- Suction Recirculation: at part load operation, the impeller's inlet of the pump starts to recirculate due to adverse pressure gradient. The reverse flow on the impeller's inlet can

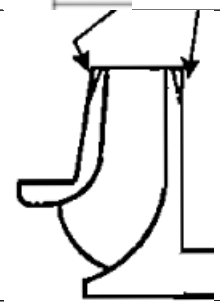
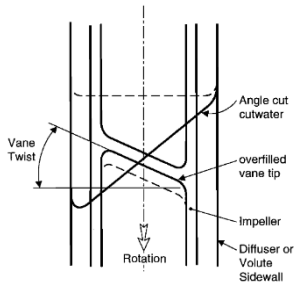
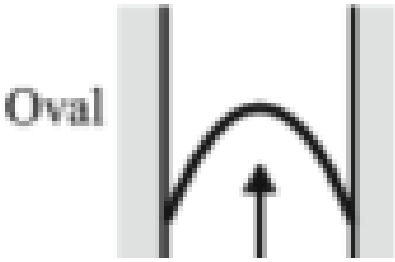
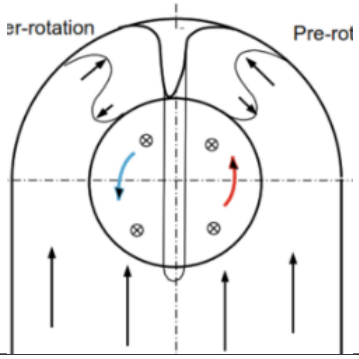
interact with the upstream static parts and reverse pressure pulsation with possible noise and a Net Positive Suction Head (NPSH) problem can occur.

- Rotor stator gap: pressure pulsation magnitude decreases by increasing the gap between the impeller TE and diffuser LE or volute tongues due to mixing effect.
- Rotor stator vane combination: the interaction between rotor and stator vanes potentially affects resonance phenomena. Using the right combination of impeller and diffuser/tongues helps to minimize vibrations. [1] calculates a m coefficient putting in relation the pulsations harmonic of the impeller and diffuser blades (or volute tongues).
- Staggering of multistage impellers: For multistage pumps staggering on the pitchwise direction, the impellers help to spread the pressure pulsations by moving it on higher frequencies.
- Staggering of double suction impeller: For double suctions pumps, the staggering angle of the blades in the pitchwise direction help to move the vane pass frequency of the pulsation towards higher values.
- Energy level of the pump: High energy pumps (pumps with high head and absorbed power) have bigger pressure pulsations. The *API 610* standard [2] recommends a minimum value of radial gap between rotor and stator to mitigate the pressure pulsation effects.

5. Design guidelines for lowering pressure pulsations

The proposed design guidelines are listed in the table below and refers to [1] [2] and [3].

Gap between diffuser and impeller	For low energy pumps (Hst<100m) D3/D2>1.015 For high energy pumps (Hst>100m and P>225kW [2]) Max between: D3/D2>1.03 and D3/D2>1.015+0.08*(1E-06*r*Hst-0.1)^0.8 if Nq<40 D3/D2>1.04+0.001*(Nq-40) if Nq>40
Gap between volute and impeller	For low energy pumps (Hst<100m) D3/D2>1.03 For high energy pumps (Hst>100m and P>225kW [2]) Max between: D3/D2>1.06 and D3/D2>1.03+0.1*Nq/40+0.07*1E-6*r*Hst
Impeller and diffuser/tongues vanes combination	M=|vx*Zimp-Vx*Zdiff| M=0 must be avoided if m=1 ask to R&D department
Zimp	Zimp*vx must avoid the acoustic resonance of crossover on BB3 pumps If H>100m, then Zimp >5 to reduce blade loading.
TE shape	Use overfilling for high risk of pressure pulsation
Rake angle	 Use rake angle <90°, opposite to cutwater oblique cutting.
Staggering double suction blades	 For high energy pumps (H.st>200m and Pabs.st>225kW) use staggered solution with pitchwise offset = 1/3 blade passage.

Oblique cut of D2	It helps to increase the D3/D2 gap.
Reduce shroud thickness	 It reduces wake generated from the external disks.
Shape of volute cutwater	  eps >35°
Throat deceleration	C3q/C2 >0.8 at BEP of the pump to avoid pressure pulsation at part load.
Inlet casing	 Rib of the casing must be overfilled to reduce wake effect. Area progression in the anular part must be designed to avoid shadding vortex formations.
Impeller inlet design	Reducing the impeller's eye help to onset the suction recirculation effects towards lower flow rates.

6. Limitations

The criteria are applicable for minimizing the pressure pulsations on centrifugal pumps.

Vibrations due to resonance phenomena of external piping or sump intake unevenness are not considered in this procedure.

The mitigation of the pressure pulsation discussed in this procedure is effective only if no resonance of the structure occurs. Under these circumstances, a deep investigation by means of advanced Finite Element Method (FEM) analysis must be performed to properly evaluate the accuracy of the redesign.

Plant operative conditions must be carefully considered before applying any modification of the design to improve pressure pulsations. In case of severe part load or overload operations, the improvement on pressure pulsation achieved at the BEP with a redesign of the impeller can be negligible.

7. References

[1] J.F.Gulich "Centrifugal Pumps" Fourth Edition - 2020

[2] API 610 12th edition

[3] Vane Pass Vibration – Source, Assessment and Correction – A Practical Guide for Centrifugal Pumps – F.L. Robinett, J.F. Gulich, T. Kaiser.

8. Attachment – Reduction of pressure pulsation on BB1 pump (job P30612)

A BB1 pump with double volute and 5 blades symmetric impeller experienced Blade Passing Frequency (BPF) at rated point with high level of vibrations.

Pump rated point:

Q	***H***	***n***	***P***	***Qbep***
[m3/h]	*[m]*	*[rpm]*	*[kW]*	*[m3/h]*
5000	*95*	*746*	*1420*	*6000*

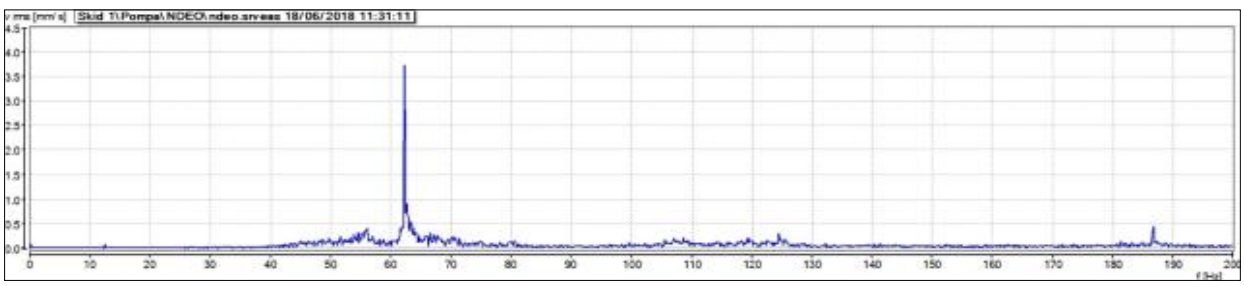


Figure 2: FFT signal on NDE support

CFD analysis figured out suction recirculation problems of the pump at the rated point due to a large impeller's eye.

A new redesign of the impeller with a reduced inlet diameter and staggered by 1/3 of the pitch helped to solve the vibration problem.

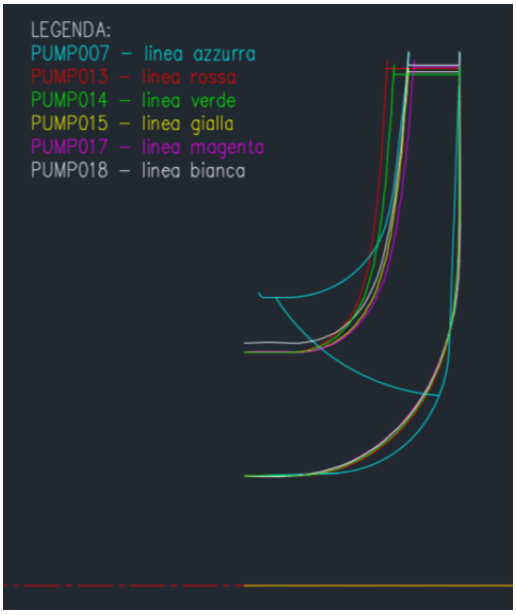


Figure 3: Meridian channel comparison NEW (Light blue) vs ORI (white)

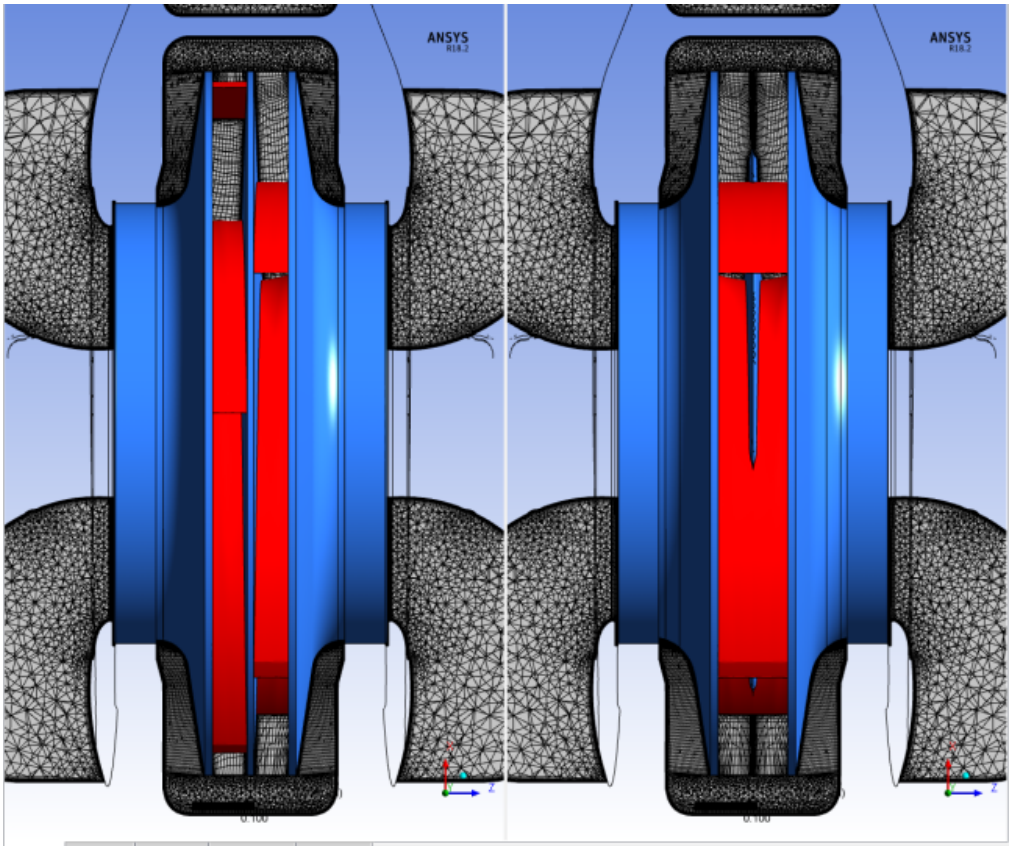


Figure 4: impeller modification NEW (left) vs ORI (right)

Q [m³/h]	NDE$_h$ [mm/s$_{rms}$]	NDE$_v$ [mm/s$_{rms}$]	NDE$_a$ [mm/s$_{rms}$]	DE$_h$ [mm/s$_{rms}$]	DE$_v$ [mm/s$_{rms}$]	note	
1500.5	8.97	12.31	5.12	7.89	9.23	minimum point	**NC30612.01.02-19 girante originale**
2508.8	6.89	10.55	4.36	5.72	8.11	middle point	
3460.2	11.86	8.71	5.44	6.84	8.14	preferred to allowable	
4996.3	10.10	9.07	4.07	5.84	7.76	rated	
5725.5	10.01	9.46	4.21	6.56	7.31	maximum allowable	
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1504	4.46	4.19	2.11	3.5	3.74	min flow	**NC30612.01.02_02-20 girante staggered diametro max**
2511	4.48	5.08	1.95	3.67	3.46		
3434	4.66	3.34	2.29	3.26	3.73	preferred to allowable	
5000	4.82	4	2.39	3.92	4.34	rated	

Figure 5: comparison of RMS vibrations Original Vs Redesign